

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Faculty of Technology and Engineering

M. Tech. (CE) Sem. – II University Examination, 2010-11

CE716 Mobile Computing and Wireless Network

Date: 20.05.2011, Friday Time: 10:00 a.m. To 01:00 p.m. Maximum Marks: 70

Instructions:

- (1) All questions are compulsory.
- (2) Indicate clearly, the option you attempt along with its respective question Number.
- (3) Figures to right indicate marks.
- (4) Rough work is done in the last page of main supplementary; don't write anything on the question paper.

SECTION-I

Q-1 Answer the following questions. [11]

- [A] What are the different types of control available in Symbian OS? [2]
- [B] What are the capability-set to define standards of Intelligent Network? [2]
- [C] What is J2ME MIDP? [2]
- [D] What is number portability in mobile communication? How is mobile number portability different from telephone portability? [2]
- [E] In CDMA, what do you mean by Pilot channel, Sync channel and paging channel? [3]

Q-2 Answer the following questions. [12]

- [A] Describe the symbian OS architecture. What are the functions of different layers in this architecture? [4]
- [B] By taking suitable example, Explain how Walsh function is usefull in wireless communication? [8]

OR

- [B] Describe the life cycle of J2ME application. Explain Generic Connection Framework. [8]

Q-3 Answer the following questions.(Attempt any Three) [12]

- [A] Draw and explain IS-95 architecture and functionalities of its elements.
- [B] Write a short note on: IN conceptual Model
- [C] Capacity of CDMA depends on criteria. What are those criteria? How does it impact capacity of CDMA?
- [D] Explain following terms:
 - 1) PARLAY
 - 2) JAIN

SECTION-II

Q-4 Answer the following questions. [11]

- [A] What are the types of attacks? [2]
- [B] What are the differences between security algorithm and protocols? [2]
- [C] What is Mobile Virtual Private Network? [2]
- [D] What are the entities of Mobile IP Network? [2]
- [E] What are SIFS, PIFS and DIFS in wireless LAN? [3]

Q-5 Answer the following questions. [12]

[A] What are the advantages and disadvantages of infra-red and radio transmission? [5]

[B] What are the components of information security? Explain each component in detail. [7]

OR

[B] Why there is need for Mobile IP. By taking suitable diagram, Explain the process of Mobile IP. [7]

Q-6 Answer the following questions.(Any Three) [12]

[A] What are the reasons for development of Wireless ATM?

[B] Explain different layers in Windows CE architecture.

[C] Explain handover mechanism in Wireless ATM.

[D] Explain Following Terms

1) DES

2) 3DES

3) AES

4) Skipjack